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An optimized hybrid CNN–Bi-LSTM framework using ACO–WOA for crop yield prediction

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Abstract

Crop yield prediction is one of the key factors that need to be considered for sustainable agriculture and food security due to climate change and increasing population. A reliable prediction model is required to determine how much yield will be produced since climate variability exacerbates global food scarcity. Existing methodologies do not account for sequential relationships and complex interactions existing in agricultural data. Moreover, another drawback with traditional methods is that their generalization is low, and hyperparameter optimization is limited, which results in inefficient models. Therefore, this study proposes a hybrid predictive method to overcome the mentioned problems. It includes 1D CNN and BiLSTM branches optimized with the Ant Colony Optimization (ACO) and Whale Optimization Algorithm (WOA), respectively. Both branches operate on the input time series simultaneously. What makes this research unique is the fact that previously there has been no research done where the CNN and BiLSTM branches have been optimized separately using different metaheuristic approaches in the agroclimatic yield prediction scenario. The CNN helps achieve effective feature extraction, whereas the BiLSTM models capture bidirectional temporal relations. The separately optimized features are aggregated to form the final predictions. Benchmarks for statistical accuracy are used to evaluate the performance of the proposed model compared to various baseline models, including the CNN–BiLSTM model with and without attention mechanisms and optimization algorithms. With RMSE = 132.1194, MAE = 104.3635, MAE ± SD of 104.3635 ± 81.0172, and R2 = 0.9678, the proposed attention-based metaheuristic CNN–BiLSTM model has produced satisfactory results. These findings show improved performance over baseline models and competitive results against existing hybrid and attention-based methods.

Keywords Hybrid CNN–BiLSTM, Metaheuristic algorithms, ACO–WOA, Crop yield prediction

1 Introduction

Agriculture is the most dominant economic activity practiced in Rajasthan; more than 60% of people in Rajasthan earn their living through agriculture. Some of the major obstacles faced in agriculture are erratic rainfall, temperature variations, soil infertility, and the occurrence of droughts [1]. The fast change in climate conditions influences

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crop growing cycles as well as production yield. This suggests that conventional agricultural systems are not adequate for effective yield planning [2]. Utilization of modern computing methods in agriculture becomes essential to enhance the accuracy of predictions and make decisions based on data analysis [3]. The revolution that uses data-based technologies is known as precision agriculture. Yield prediction of crops becomes the most crucial factor in precision agriculture. This could assist in irrigation planning, fertilizer distribution, planting time, and land utilization [4]. The early estimation of yield is helpful for the farmers in planning better harvesting times and managing the supply of products. Furthermore, proper yield estimation may assist in formulating suitable agricultural policy by government bodies by utilizing available resources wisely and adopting necessary measures for risk mitigation under adverse climatic situations [5]. Rajasthan relies on monsoonal irrigation and faces adverse climatic situations frequently; therefore, it needs an advanced prediction system [6]. Traditional statistical approaches lack the ability to capture nonlinearity and interaction between climatic, soil, and agronomical factors [7]. With the availability of increasing agro-climatic and remote sensing data, machine learning (ML) and deep learning (DL) approaches can be considered effective ways of modeling nonlinear and multivariate relations. Time series forecasting plays an important role in agricultural predictions, taking into account that crop production happens sequentially and is affected by time-based parameters. Weather parameters such as rainfall, temperature, humidity, and solar radiation vary over time depending on the different stages of crop growth. Traditional algorithms assume independence between data without taking into account their relation to time. Therefore, they cannot account for the presence of nonlinear and temporal relationships that exist in agricultural data. Deep learning models can learn from complex patterns, but a deep learning algorithm alone might not be sufficient enough to learn all the characteristics of crop production. CNNs have proven useful in extracting features from local and hierarchical levels for data with a structure such as time-series windows [8]. On the contrary, RNNs and their variations such as Long Short-Term Memory (LSTM), have been designed to detect long-term dependency features, which can be helpful in forecasting yields [9]. Yet, it is possible that RNN and LSTM might fail to capture short-term dependencies. Hence, a novel idea of hybrid deep learning algorithms combining both CNN and BiLSTM came into existence. The introduction of this new paradigm aids in developing more intricate representations for the model, thus improving accuracy. The CNN layer contributes to making sequential models more expressive by incorporating transient effects and interactions between features. Conversely, BiLSTM networks evaluate sequences in both forward and backward directions. This allows the model to obtain information regarding both the past and the future simultaneously [10]. This two-way learning enhances prediction, especially when crop yield depends on seasonal trends that accumulate over time. Recent advancements show that adding an attention mechanism enhances predictive accuracy. The attention layer assigns varying weights to different time steps that change over time. This enables the model to focus on the most important information when making predictions [11]. The attention-based CNN–BiLSTM model performs well, but its efficiency depends on the selection of appropriate hyperparameters. Manual tuning and grid search require a lot of resources and do not always identify globally optimal solutions. Therefore, there is a need to develop an attention-based metaheuristic DL framework that can learn spatial–temporal patterns from

multi-source agroclimatic data while automatically tuning hyperparameters to improve prediction accuracy. One of the significant limitations present in existing model structures is that most studies employ traditional optimization techniques like random search and Bayesian optimization. In addition, a single optimization technique is used for the tuning of all architectural elements. This is because each component of the model uses different parameters and learning dynamics. For instance, the CNN branch uses parameters that are predominantly discrete, including filter numbers and kernel arrangements, which require efficient combinatorial search. On the other hand, BiLSTM requires continuous parameters such as learning rate, dropout rate, and number of hidden units that require smooth global optimization. The use of one optimization for both parts can be inefficient, as the approach may not adequately cover either the discrete or the continuous parameters. Therefore, this research will propose an attention-based metaheuristic model that will independently optimize both the CNN and BiLSTM branches using two different optimization algorithms. The CNN branch is optimized using Ant Colony Optimization (ACO), a swarm intelligence technique inspired by how ants search for food. It adopts the probability search strategy of pheromones to efficiently traverse through the solution space while avoiding the problem of getting stuck in local optima [12]. On the other hand, the BiLSTM module is optimized using the Whale Optimization Algorithm (WOA). WOA has been developed based on the hunting strategy of the humpback whale using bubble nets. It possesses excellent global searching ability along with quick convergence [13]. WOA is more suitable for solving continuous optimization problems that involve fine-tuning of parameters. The ability of WOA to dynamically balance the exploration–exploitation ratio is advantageous in optimizing hyperparameters of BiLSTM. This is achieved by matching each optimization technique with the underlying characteristics of the corresponding component within the model, resulting in a faster process of parameter adjustments, higher convergence rates, and better feature learning capabilities. This contributes to an increase in predictive accuracy beyond that of traditional models using single optimization strategies. This study is consistent with the United Nations' Sustainable Development Goal 2 in that it utilizes state-of-the-art deep learning and optimization techniques to improve crop yield prediction in an attempt to ensure sustainable agriculture. Recent research has significantly enhanced the prediction accuracy of crop yield using attention-based CNN–BiLSTM model with optimization techniques. However, there are still a few limitations to these models. The study by Li et al. [14] shows the efficiency of the attention-LSTM model. However, the model lacks an optimization technique, making the model less optimal in terms of tuning the model parameters. Similarly, in the study by Sun et al. [15], the WT-CNN–BiLSTM model was developed to efficiently learn both spatial and temporal features. But the model lacks an optimization technique to reach optimal performance. In addition, recent models developed by Zhang [3] have been effective in improving accuracy through optimization techniques. However, the model relies on a single optimization technique, making the model less effective in terms of balancing exploration and exploitation. Similarly, Song et al. [16] developed the TPCLA model using transfer learning and attention techniques. The model was effective in terms of accuracy; however, the model relies on a single optimization technique, making the model less effective in terms of balancing exploration and exploitation. The model also lacks independent optimization techniques for CNN and LSTM, which might affect the learning of optimal features.

In addition, these studies have demonstrated the effectiveness of hybrid machine learning frameworks combined with metaheuristic optimization across various domains. For example, optimization-based hybrid methodologies have been successfully implemented in applications including medical image classification by Özkan et al. [17], plant disease detection in real-time by Yağ et al. [18], and time-series forecasting based on LSTM models with improved optimization algorithms by Karasu et al. [19]. In addition, the most recent works by El-Kenawy et al. [20], El-Kenawy et al. [21], Radwan et al. [22], and Alhussan et al. [23] show the increasing significance of combining optimization algorithms and deep learning algorithms to solve various problems, such as forecasting renewable energy, predicting smart city electricity, and network optimization.

Such research highlights the importance of leveraging domain-specific feature learning together with adaptation optimization methods. Nevertheless, current literature primarily centers around single-domain applications that depend solely on a single optimization algorithm or a common optimization algorithm to adapt both spatial and temporal features without distinguishing them. Moreover, there is a lack of studies involving the implementation of these optimal models in time-series forecasting in agriculture. From these studies, some of the key research gaps are highlighted below. Firstly, most studies concentrate mainly on spatial information obtained from satellite imagery or temporal information from climatic time-series data but not on their combination. Second, although attention-based CNN–BiLSTM models have been explored, most studies do not apply metaheuristic optimization to tune the architecture and hyperparameters, which can significantly affect performance. Third, existing approaches generally rely on single optimization strategies or no optimization at all, and there is limited research on independent optimization of CNN and BiLSTM components using different metaheuristic algorithms. Finally, relatively few studies utilize long-term agroclimatic and soil nutrient datasets for large agricultural regions such as Rajasthan, limiting their practical applicability. To address these gaps, the present study extends the concept of optimization-driven modeling to crop yield prediction by introducing an attention-based CNN–BiLSTM model with metaheuristic optimization techniques, namely ACO-WOA, as depicted in Fig. 1. Unlike the existing models, the proposed model can optimize the spatial and temporal components independently, thus making the model more effective in terms of feature extraction. The combination of these two optimization techniques avoids the limitations faced in the existing models, making the proposed model more effective in terms of prediction accuracy. Hence, the proposed approach provides an innovation in terms of concepts, as it overcomes the drawbacks of unified optimization techniques by implementing branch-specific optimization rather than expanding the

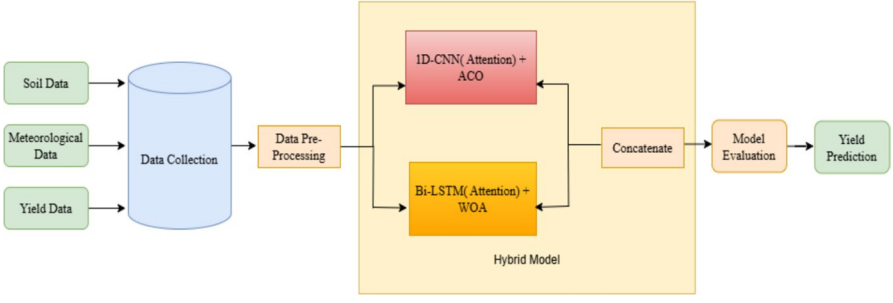


Fig. 1 Structure of attention-based metaheuristic CNN–BiLSTM model

search domain. Unlike single-optimizer approaches that apply a unified search over the entire architecture, the proposed method decomposes optimization into two specialized processes tailored to heterogeneous parameter spaces. This reduces search interference, improves convergence efficiency, and enhances the balance between exploration and exploitation.

1.1 Motivation

It is challenging to predict crop yields accurately because of the complicated interactions between climate variables, soil conditions, and changing environmental patterns over time that affect agriculture. Traditional statistics or standalone deep learning approaches have been adopted by many modern prediction techniques, but these techniques lack the ability to analyze both temporal and spatial relationships present in most agroclimatic datasets.

Furthermore, most research focuses on optimizing the overall deep learning model using only a single tuning method, ignoring the fact that different parts of hybrid architectures have their own learning characteristics that may need to be optimized separately. Because of these issues, a more advanced prediction framework is needed to effectively identify complex feature correlations, model temporal dependencies, and make the architecture's different parts work better in a more flexible way. Motivated by these challenges, this study presents a parallel CNN–BiLSTM with attention framework improved by dual metaheuristic optimization to improve the accuracy of crop yield predictions using district-level agro-climatic data from large meteorological and agricultural datasets.

1.2 Novel contribution of this article

This study proposes an attention-based meteheuristic CNN–BiLSTM model for district-level wheat yield prediction in Rajasthan, India. The main contributions are as follows:

a. A multi-district agroclimatic panel dataset (1981–2017) is created by integrating meteorological variables from NASA POWER and soil nutrient and yield data from ICRISAT. The panel formulation amplifies effective sample size and improves statistical robustness.
b. A hybrid deep architecture is proposed, where the CNN module is responsible for localizing inter-feature interactions, and the BiLSTM module is responsible for representing long-term temporal correlations using a fixed sliding window structure. The parallel architecture ensures the complementary nature of spatial–temporal feature extraction.
c. This study proposes to optimize CNN's hyperparameters using ACO and BiLSTM's hyperparameters using WOA separately, unlike previous works that adopted a single global optimization technique. The branch-specific optimization technique allows for heterogeneous search dynamics to be adopted for convolutional and recurrent components.

1.3 Organization

The rest of the paper is organized as follows: Sect. 2 present the literature review. Section 3 introduces the study area and dataset used in the study. Section 4 provides detailed explanations of the methods, and in Sect. 5, the experimental setup and ablation

study are presented. The results and discussion are illustrated in Sect. 6. Performance evaluation and comparative analysis with base research are described in Sect. 7, and Sect. 8 concludes the paper.

2 Literature review

The goal of the literature review regarding crop yield prediction is to understand and evaluate the vast amount of knowledge that currently exists regarding crop yield prediction [3]. It is a crucial part of agricultural research. Table 1 has demonstrated the effectiveness of optimization-driven models in improving forecasting and prediction accuracy in yield prediction. Varghese et al. [24] introduced a hybrid CNN-RNN model for agricultural yield prediction that adeptly integrates spatial feature extraction (CNN) with temporal dependency modeling (LSTM-based RNN). Experimental findings indicate that the suggested model surpasses the ANN, attaining superior accuracy (up to 88%), precision (89%), recall (85%), and F-measure (86%) from 2018 to 2021. Bharti et al. [25] proposed a hybrid CNN–LSTM model in which CNN extracts features from agro-climatic data, while LSTM captures temporal connections. The model has a high accuracy with a high R^2 of 0.97, as well as low error measurements of 88.05 RMSE and 65.70 MAE. The study uses the hybrid deep learning models proposed by Oikonomidis et al. [26] and compares the results with the XGBoost method for predicting soybean production, using large quantities of weather, soil, and management information from the Corn Belt region of the United States. The hybrid CNN-DNN model exhibited the superior performance, with RMSE = 0.266, MAE = 0.199, and $R^2 = 0.87$. A novel methodology that incorporates Z-score cleaning, CEEMDAN-SE reconstruction, CNN feature extraction, and BiLSTM modeling was proposed by Li et al. [27]. The proposed CEEMDAN-SE-CNN–BiLSTM model has shown exceptional performance in prediction with RMSE = 17.69, MAE = 12.47, and $R^2 = 0.9817$. The CNN–BiLSTM-CYP model was introduced by Saini et al. [28], combining convolutional layers to extract features in time series data with bidirectional LSTM to consider forward and backward interactions in time series data. The CNN–BiLSTM_CYP model was found to perform better than other conventional approaches such as ARIMA, GPR, HW, and Stacked LSTM by significantly reducing RMSE and MSE values. The major findings of this study showed that the CNN–BiLSTM_CYP approach had better performance in terms of RMSE (4.05) and MSE (16.40) compared to other approaches. In the work by Reddy et al. [29], the researchers develop an advanced hybrid deep learning model that improves temporal modeling, feature selection, and reduce prediction errors. Deep learning techniques like CNN, LSTM, and RNN significantly improve the accuracy of crop yield prediction. Even further accuracy can be achieved by using optimization techniques like ICSO-RNN. In this case, the accuracy can be up to $R^2 = 0.99$. A Weight Agnostic Neural Network (WANN), proposed by Jovanovic et al. [30] was based on the concept of a metaheuristic approach for predicting agricultural yield. The main objective is to enhance the precision of the prediction by using a collaborative dual-layer approach. In the modified WANN approach, the MAE and R^2 values obtained were 0.017698 and 0.886555, respectively. In the research paper presented by Jia et al. [9], the researchers proposed the use of a hybrid CNN–BiLSTM model with an attention mechanism for the prediction of crop production and fertilizer recommendation, which can overcome the limitations of the conventional machine learning approach. The proposed methodology has an accuracy of 98%

Table 1 Comparative summary of crop yield prediction studies

Refs	Crop	Region	Data Sources	Model Type	Temporal Direction	Statistical Validation
Varghese et al. [24]	Rubber (Hevea) 3	Kerala, India	Soil, rainfall, humidity, temperature, wind speed (2018–2021), IMD & RRII data	Hybrid CNN-RNN with LSTM	Sequential temporal modeling with k-year dependency	Accuracy up to 88%, improved precision, recall, F-measure over ANN
Bharti et al. [25]	Wheat1	Punjab, India	Government yield data (1980–2018), NASA POWER weather data (temperature, rainfall, soil moisture, humidity, irradiance, area, tubewells)	CNN-LSTM Hybrid	Forward time-series modeling	Evaluated using RMSE, MAE, MSE, R^2; CNN-LSTM showed lower error and higher R^2 than LSTM
Oikonomidis et al. [26]	Soybean	9 States, USA (Corn Belt)	Weather, Soil, and Management Data (1980–2018), 395 features, 25,345 samples	Hybrid CNN-DNN (Best), CNN-RNN, CNN-LSTM, CNN-XGBoost, XGBoost	Year-based time-series modeling	Best Model (CNN-DNN): RMSE = 0.266, MSE = 0.071, MAE = 0.199, R^2 = 0.87; XGBoost (Feature Selected): R^2 = 0.83
Li et al. [27]	Sugarcane	India (region-specific study)	Historical yield data with climatic and environmental variables	CNN–BiLSTM-CYP (Hybrid Deep Learning)	Bidirectional time-series modeling (BiLSTM)	RMSE ≈ 4.05, MSE ≈ 16.40; compared with ARIMA, GPR, Holt–Winters, Stacked-LSTM
Saini et al. [28]	Wind Power (Energy Forecasting Study)	Hubei, China	Wind power time-series data with meteorological preprocessing (CEEMDAN + SE)	CEEMDAN-SE-CNN–BiLSTM	Bidirectional time-series modeling with signal decomposition	RMSE = 17.69, MAE = 12.47, R^2 = 0.9817; outperformed BiLSTM and CNN-LSTM
Reddy et al. [29]	Mixed Crops (Nuts)	Andhra Pradesh, India	Monthly yield data (2011–2021), soil, temperature, water, solar radiation, area, production	ICSO-RNN (Inertia-based Cat Swarm Optimization + RNN)	Time-series sequential modeling	R^2 = 0.99, MAE = 0.77, MSE = 0.68, RMSE = 0.82
Jovanovic et al. [30]	Wild Blueberry & Multi-crop Dataset	USA (Blueberry – Maine) & Global FAO Dataset	Kaggle Blueberry dataset + FAO Crop Yield dataset (weather, rainfall, temperature, pesticides)	Metaheuristic-tuned WANN (MSRSA-WANN, Two-Layer Framework)	Year-based time-series forecasting	MAE = 0.017698; R^2 = 0.886555; Evaluated using MSE, RMSE, MAE, R^2 with statistical validation & SAGE analysis
Jia et al. [9]	31 Crops (Rice, Wheat, Maize, etc.)	India (GitHub agricultural dataset)	Crop dataset (Temperature, Humidity, pH, Rainfall) + Fertilizer dataset (N, P, K, pH)	CNN–BiLSTM with Attention (Hybrid Deep Learning Ensemble)	Bidirectional time-series modeling (BiLSTM)	Crop Dataset: 98% Accuracy, F1 = 0.98; Fertilizer Dataset: 99% Accuracy, F1 = 0.99; Lowest MSE = 0.04 (crop), 0.02 (fertilizer)
Zhang et al.[3]	Winter Wheat	China	Multisource remote sensing + agro-climatic data	Bayesian Optimization-based CNN–BiLSTM	Multivariate Time-Series	RMSE = 616.99, MRE = 7.14%, R^2 = 0.81

Table 1 (continued)

Refs	Crop	Region	Data Sources	Model Type	Temporal Direction	Statistical Validation
Fariz et al. [31]	Cassava, Maize, Rice, Wheat, Sorghum, Soybean, Sweet Potato, Yam, Plantains, Potatoes	102 Countries (Global)	FAOSTAT (2000–2013): Yield, Rainfall, Pesticide Use, Temperature, Crop Type, Nation, Year	Hybrid ANN-COA (Levy Flight Enhanced Coati Optimization + ANN)	Time-series with year-based prediction(2000–2012 → 2013)	RMSE = 14,338.6; MAE = 8685.3; MAPE = 0.06803; R^2 = 0.97296; Outperformed ANN-PSO, ANN-GWO, ANN-WOA, LSTM, CNN-RNN, ARIMA
Li et al. [32]	Wheat, Rice, Maize	Wheat599 & Wheat2000 (CIMMYT/Iran), Rice299 (IRRI), G2F_2017 (USA)	SNP markers (1,279–73,147 SNPs) + multiple phenotypic traits	Hybrid DL: CNN-LSTM, CNN-ResNet, LSTM-ResNet (Best), CNN-ResNet-LSTM	Sequential + Residual learning (LSTM + ResNet skip connections)	tenfold CV; Pearson r up to 0.946 (DTP), 0.897 (PH), 0.747 (GW), outperforming BayesA/B/C & BRR
Li et al. [14]	Rice	Qian Gorlos County, Northeast China	Sentinel-2 multispectral imagery (NDRE, GCVI, NDWI) + ground-measured yield data (2022–2023)	Attention-based LSTM (ALSTM) with LASSO, RF, LSTM as baselines	Time-series (GP1–GP4; forward sequential modeling with attention weighting)	R^2 = 0.88, RMSE = 341.82 kg/ha, MAE = 280.29 kg/ha; fivefold cross-validation; < 2% deviation from official statistics
Sun et al. [15]	Crop yield (greenhouse crops)	Controlled greenhouse environment	Environmental sensor data	CNN, BiLSTM, WT-CNN–BiLSTM	Sequential time-series	RMSE, R^2 WT-CNN–BiLSTM: $R^2 \approx 0.95$
Song et al. [16]	Wheat	China	Govt. statistical yearbook(1993–2024) climatic + social-economic data	Transfer learning based parallel CNN-LSTM-attention	Multivariate time-series	RMSE = 0.394, MAE = 0.326, R2 = 0.904
Rufaioglu, et al.[33]	Wheat	Agricultural fields	UAV imagery + climatic variables	CNN variants	Mainly spatial features	RMSE, R^2 CNN: $R^2 \approx 0.98$, RMSE ≈ 0.12
Arya et al. [34]	Bananas, Rice, Wheat, Soybeans, Maize, Cassava, etc	Global (UNFAO, Our World in Data)	Annual Time-Series Yield Data (tonnes/hectare)	ARIMA, ETS, ANN, LSTM, PLS, EMD-ANN, Wavelet-ANN, ETS-ANN	Long-term time-series forecasting (20-year horizon)	Best Models: PLS (MAE = 0.142), ETS-ANN (R^2 up to 0.809); Metrics: MAE, RMSE, R^2, MAPE, CDVI

for crop prediction and 99% for fertilizer recommendation, proving its high robustness. The purpose of the research proposed by Zhang et al. [3] is to improve the estimation of winter wheat yield. This study utilizes a Bayesian optimization-based CNN–BiLSTM approach for integrating multisource remote sensing and agroclimatic information. The model showed better results in terms of prediction accuracy (RMSE ≈ 616.99, MRE ≈ 7.14%, $R^2 \approx 0.81$). Fariz et al. [31] proposes a hybrid model of ANN-COA (Artificial Neural Network with Lévy flight-enhanced Coati Optimization Algorithm) for the enhancement of global crop yield forecasting in 102 countries. This is done to enhance the performance of the ANN model through the optimization of the weights and biases

using the COA with the aid of the Levy flight algorithm. The suggested ANN-COA demonstrated enhanced accuracy (RMSE = 14,338.6; MAE = 8685.3; MAPE = 0.06803; R^2 = 0.97296), surpassing ANN-PSO, ANN-GWO, ANN-WOA, LSTM, CNN-RNN, and ARIMA. Li et al. [32] have suggested developing hybrid deep learning algorithms to increase the accuracy of genomic prediction in crop breeding. The main aim is to incorporate spatial feature learning using CNNs, sequence learning using LSTMs, and residual learning using ResNet to overcome the limitations of conventional statistical-based genome selection techniques. The models have been able to achieve maximum correlation coefficients of 0.946 for DTP and 0.897 for PH, signifying high robustness and accuracy compared to Bayesian-based and individual deep learning-based techniques. ALSTM architecture for pixel-scale early rice yield estimation was proposed by Li et al. [14] that utilized a combination of long short-term memory networks and an attention mechanism to extract relevant growth stage information from Sentinel-2 vegetation indices. The study suggests that the heading-milk ripening stage (GP4) is the most effective early prediction period for rice yield estimation with an R^2 value of 0.88, RMSE of 341.82 kg/ha, and MAE of 280.29 kg/ha compared to other techniques such as LASSO, RF, and LSTM. The paper by Sun et al. [15] attempts to enhance the accuracy of crop yield prediction in complex environments through the application of deep learning techniques. A WT-CNN–BiLSTM hybrid model has been proposed, which uses wavelet convolution, residual CNN feature extraction, and Bidirectional LSTM for the prediction of crop yield. The proposed model has resulted in $R^2 \approx 0.92$ and RMSE ≈ 9.68 g, with higher accuracy than other models like CNN–LSTM and CNN–BiLSTM. Transfer Learning-based Parallel CNN-LSTM-Attention (TPCLA) is proposed by song et al. [16], which combines CNN and LSTM for feature learning and temporal dependency learning. This model is augmented by attention mechanism to emphasize significant features. The model demonstrates superior predictive performance with an RMSE of 0.394, MAE of 0.326, and R2 of 0.904, surpassing traditional RNN, LSTM, and hybrid models. The paper presented by Rufaioglu et al. [33] emphasizes the enhancement of wheat yield prediction by integrating diverse agricultural data sources. The research goal is to improve the accuracy of the predictive model by integrating high-resolution images obtained from the UAV and ground sensors at different stages of crop growth. The research utilized the Convolutional Neural Network (CNN) for feature extraction from the images obtained from the UAV and then combined the images with the environmental sensors for yield estimation, which proved to have a high level of accuracy with R^2 close to 0.98. In the study by Arya et al. [34], the authors propose a hybrid multi-model machine learning framework for the prediction of the yield of staple crops. The intention of the study is to improve the accuracy of long-term forecasts of crop yields by addressing the challenges of nonlinearity, non-stationarity, and volatility in the data on crop yields globally. The study found that the hybrid models using PLS and ETS had the best performance in the prediction of Cassava yield (R^2 = 0.809) and Rice yield (R^2 = 0.762), while the performance of the ANN and decomposition model was low.

3 Material

Time series models facilitate enhanced predictions by identifying critical aspects within the time-dependent sequential relationships of the data. The capabilities of DL have emerged as an important factor in the agriculture field. This study demonstrates that

CNN and BiLSTM models employed in time series analysis may comprehend intricate and long-term connections in the data, whereas basic linear models are inadequate in capturing these patterns.

3.1 Study area

India's largest state in terms of land is Rajasthan. The climate of Rajasthan is primarily arid and semi-arid. It is located in the country's northwest and is distinguished by high temperature variability, frequent droughts, and little to no rainfall [35]. Annual rainfall varies from 100 to 450 mm within the arid and semi-arid areas of Rajasthan. The state has varied agroclimatic zones, which range from the Thar Desert to irrigated regions close to the Indo-Gangetic plains [36]. Pulses, wheat, barley, mustard, and pearl millet are important crops. Rajasthan is highly vulnerable to climate change and water scarcity because of its unpredictable climate and reliance on groundwater and monsoon rains.

3.2 Dataset

The datasets utilized in this study consist of district-wise agroclimatic and soil parameters with varying temporal resolutions in Rajasthan [35]. The dataset covers the time period from 1981 to 2017, which contains 12 input features, and wheat yield is the target variable, which represents the crop yield output. The annual district-wise wheat yield and soil nutrient data, including nitrogen (N), phosphorus (P), potassium (K), and soil pH, are obtained from ICRISAT [37]. Monthly agroclimatic variables—precipitation, maximum and minimum temperature, relative humidity, soil wetness, and maximum and minimum wind speed—are retrieved from NASA POWER using the centroid coordinates of each district. Since wheat is cultivated during the Rabi season, climate variables influencing yield are derived by aggregating monthly data from October (Y – 1) to March (Y) corresponding to each yield year. Specifically, total seasonal rainfall and mean seasonal values of temperature, humidity, and wind speed are computed to represent crop-stage-specific environmental conditions. The seasonal climate variables are merged with district-wise annual soil and yield data using district and year as alignment keys, resulting in a balanced spatio-temporal panel dataset comprising 33 districts over 37 years (1221 observations). The data is structured such that it follows the format of district-level panels. This means that each district has its own unique features when it comes to agroclimatic and soil properties. Due to inter-district variation, the sliding window approach will be performed for each individual district so as to retain the time-series relationship without combining information from different districts. This makes it possible for the model to indirectly learn the unique patterns over time that may be present in each district using the technique of sequence modeling. The identity of the district is not used as one of the features, but the model learns about the heterogeneity in space using the differences among the input variables for each district. This makes each sequence unique based on its environment, enabling the model to learn region-specific dynamics indirectly. To capture long-term temporal dependencies, a sliding window approach with a time step of $T = 7$ years is applied independently within each district, generating 990 spatio-temporal training sequences. To maintain consistency of data and stability of the models, certain preprocessing steps have been taken. Missing values in the dataset are filled by using mean imputation on a per-column basis. This approach preserves the overall distribution of the data while avoiding information loss.

Outliers are not handled by any specific outlier processing step. Rather, the effect of outliers on the analysis is indirectly addressed via min–max normalization, where each feature is rescaled in the interval [0,1]. This step is crucial for deep learning algorithms because it guarantees the same importance of features and avoids gradient problems. Further feature transformations like logarithmic scaling are not performed since the min–max normalization and deep learning model can deal with skewness and non-linearity of data distribution on their own. 70% of the dataset is divided into training, 15% validation, and 15% testing subsets. Random shuffling is avoided, and the dataset is partitioned chronologically to preserve the temporal structure and prevent information leakage from the time-series data. Cross-validation is not applied to avoid temporal data leakage and overly optimistic performance estimates. Instead, a chronological hold-out strategy is adopted to ensure realistic forecasting conditions. Walk-forward validation is not applied due to its high computational cost. Regularization methods help in reducing overfitting. Consistency is maintained in all the preprocessing, design, and optimization methods discussed in this paper in all the experimental setups.

4 Methodology

4.1 Attention-based CNN–BiLSTM model

Spatial and temporal variables influence the wheat yield, and both of these are strongly connected with each other [16]. CNN is applied for extracting the spatial features since it can be effectively utilized for identifying local patterns and hierarchies. The CNN automatically learns important feature interactions among climatic and soil variables across time steps. The model can identify important spatial patterns and environmental dependencies that impact crop growth, so the CNN model is selected for crop yield prediction. It works well with BiLSTM to make the CNN–BiLSTM models. BiLSTM is selected over LSTM because it can capture temporal dependencies in both forward and backward directions within time-series data [14]. The bidirectional learning enables the model to better understand long-term temporal relationships in agroclimatic data, leading to improved feature representation and more accurate yield prediction. This model uses the spatial feature extraction of CNNs and the sequential learning abilities of LSTMs simultaneously [38]. The model combines 1D CNN layers, BiLSTM layers, and an attention mechanism to let it learn complementary representations from multivariate time-series data. The 1D CNN branch analyzes the input time-series data to extract advanced spatial–temporal features and short-term and local dependencies. It helps identify early-stage patterns like temperature variation, soil moisture, and precipitation patterns [39]. The convolution process uses sliding filters to automatically create discriminative feature maps, so there is no need for manual feature engineering. Pooling layers keep important information while reducing the spatial dimensionality. Then they send the retrieved features to dense layers. An attention mechanism is applied to the CNN outputs, selectively highlighting the most important temporal regions and feature maps, which enhances feature significance. Equation 1 is showing the input representation of the model. The multivariate time-series input is defined as:

$$\mathrm{X} = \{\mathrm{x}_1, \mathrm{x}_2, \ldots\ldots, x_\mathrm{T}\}, \quad \mathrm{x_t} N\ \mathrm{R_d} \tag{1}$$

where T is the time step (T = 7); d is the number of features; $\mathrm{x_t}$ is the feature vector at time step t.

The CNN feature extraction is represented in Eq. 2. The 1D convolution operation is defined as:

$$c_t^{(k)} = \sigma \sum_{i=0}^{k=1} W_i^{(k)}.x_{t+1} + b^{(k)} \tag{2}$$

where $W_i^{(k)}$ is the weight of the ith element in the kth convolution filter; k is the kernel size; $b^{(k)}$ is the bias; σ is the activation function (ReLU); $c_t^{(k)}$ is the output feature at time t for filter k.

The CNN output feature map is represented in Eq. 3.

$$\mathrm{C} = \{\mathrm{c}_1, \mathrm{c}_2, \ldots\ldots \mathrm{c}_\mathrm{T}\} \tag{3}$$

CNN Attention mechanism is showing from Eq. (4) to (6).

Attention scores:

$$e_t^{CNN} = v^T \ \tanh\left(W_c c_t + b_c\right) \tag{4}$$

Attention weights:

$$a_t^{CNN} = \frac{\exp\left(e_t^{CNN}\right)}{\sum_{i=1}^{T} \exp\left(e_i^{CNN}\right)} \tag{5}$$

Attention-weighted CNN output:

$$\hat{c} = \sum_{t=1}^{T} a_t^{CNN}.c_t \tag{6}$$

where W_c is the weight for attention layer; b_c is the bias for attention layer; a_t^{CNN} is the attention weight at time t; $\hat{c}$ is the aggregated CNN feature vector.

The BiLSTM branch processes the same input sequence at the same time, but separately, to model long-term bidirectional temporal interdependence during growing seasons. The gated structure solves the vanishing gradient problems while keeping important temporal information. For BiLSTM, equations are the same as the LSTM equations [40]. For each time step t, the BiLSTM has a forward hidden state $\overrightarrow{\mathrm{h_t}}$ and a backward hidden state $\overleftarrow{\mathrm{h_t}}$. These are concatenated to form the final output at time t, as shown from Eqs. (7) to (10).

$$\mathrm{h_t^{Bi\ -\ LSTM}} = \mathrm{concat}\left(\overrightarrow{h_t}, \overleftarrow{h_t}\right)\Big) \tag{7}$$

BiLSTM attention Mechanism:

Attention scores:

$$e_t^{LSTM} = v^t \tanh\left(W_h h_t^{BiLSTM} + b_h\right) \tag{8}$$

Attention weights:

$$a_t^{LSTM} = \frac{\exp\left(e_t^{LSTM}\right)}{\sum_{i=1}^{T} \exp\left(e_i^{LSTM}\right)} \tag{9}$$

Attention-weighted BiLSTM output:

$$\hat{h} = \sum_{t=1}^{T} a_t^{LSTM} \cdot h_t^{BiLSTM} \tag{10}$$

where W_h,b_h is the attention parameters for BiLSTM; a_t^{LSTM} is the attention weights; $\widehat{h}$ is the aggregated BiLSTM feature vector.

An attention method applied to the output of BiLSTM dynamically assigns more weight to important variables and time steps that affect yield outcomes. The attention-weighted features from both the CNN and BiLSTM branches are combined, which helps the model to learn about spatial, temporal, and context-aware information that is important for making accurate predictions about agricultural yields. The output of CNN and BiLSTM branches are concatenated as shown in Eq. (11):

$$z = \left[\hat{c}; \hat{h}\right] \tag{11}$$

The predicted crop yield is computed as shown in Eq. (12):

$$\hat{y} = W_o z + b_o \tag{12}$$

where z is the fused feature; W_o is the weight matrix of output layer; b_o is the bias term; $\widehat{y}$ is the predicted crop yield.

4.2 Optimization ALGORITHM

4.2.1 ACO

ACO is a population-based metaheuristic inspired by the foraging behavior of actual ant colonies, wherein artificial ants deposit and adhere to pheromone trails to identify optimal paths through positive feedback mechanisms. ACO is applied to augment model performance by iteratively strengthening optimal solution pathways based on pheromone levels, thus enhancing classification accuracy [41]. In feature subset selection, ACO uses pheromone intensity as an indirect communication method to probabilistically lead the selection of pertinent features, with frequently selected feature routes accumulating more pheromone levels, hence enhancing their probability of subsequent selection [42]. ACO regularly exhibits its efficacy in addressing intricate optimization challenges via collective intelligence, probabilistic decision-making, and iterative pheromone reinforcement methods [43]. The probability of selecting a CNN filter configuration is shown in Eq. (13):

$$P_i = \frac{T_i^a}{\sum_{j=1}^{n} T_j^a} \tag{13}$$

where P_i is the probability of selecting the ith solution; T_i is the pheromone value associated with solution i; a is the pheromone importance factor; n is the total number of candidate solutions.

ACO Pheromone update is represented in Eq. (14) and (15):

$$\text{Pheromone evaporation: } T_i = (1-p)\, T_i \tag{14}$$

where p is the evaporation rate; T_i is the updated pheromone value after evaporation.

Pheromone reinforcement:

$$T_i = T_i + \frac{1}{RMSE_i + \in} \tag{15}$$

$RMSE_i$ is the root mean square error of the ith solution; $\in$ is the small constant to avoid division by zero; T_i is the updated pheromone value after reinforcement.

4.2.2 WOA

The WOA is a bio-inspired metaheuristic method derived from the hunting techniques and social activities of whales, especially humpback whales [31]. Inspired by the unique bubble-net feeding technique, WOA mathematically simulates prey encirclement, spiral bubble-net assaults, and position-updating tactics to efficiently investigate and utilize the search space for global optimization, as shown in the WOA algorithm. The algorithm maintains a population of candidate solutions, or whales, with each whale symbolizing a potential resolution to the optimization problem. Inspired by leadership and cooperative hunting activities in whale groups, WOA embodies a collaborative search process similar to social whale dynamics. Humpback whales exhibit exceptional intelligence and flexibility, utilizing coordinated bubble-net feeding to ensnare krill and tiny fish, which underpins WOA's spiral updating and encircling systems. WOA attains a balance between exploration and exploitation by converting biological behaviors into mathematical operators, facilitating efficient convergence to optimal solutions in complex nonlinear optimization problems [44]. The encircling prey and bubble-net attacking mechanism of the WOA model are represented by Eqs. (16) to (22):

1. Encircling prey:

$$\mathrm{D} = |\mathrm{C} \cdot \mathrm{X} * - \mathrm{X}| \tag{16}$$

$$\mathrm{X}\,(\mathrm{t}+1) = \mathrm{X} * - \mathrm{A} \cdot \mathrm{D} \tag{17}$$

WOA coefficient vectors calculated as:

$$\mathrm{A} = 2\mathrm{a} \cdot \mathrm{r} - \mathrm{a} \tag{18}$$

$$\mathrm{C} = 2 \cdot \mathrm{r} \tag{19}$$

$$a = 2 - \frac{2t}{T} \tag{20}$$

2. Bubble-net attacking:

$$\mathrm{X}\,(\mathrm{t}+1) = D' \cdot e^{bl} \cdot \cos(2\pi \mathrm{l}) + \mathrm{X} * \tag{21}$$

$$\mathrm{D}' = |\mathrm{X} * - \mathrm{X}| \tag{22}$$

where X is the whale position; X* is the best whale position; D, D′ is the distance vectors; A, C is the coefficient vectors controlling movement; a is coefficient values from (2 to 0); r is the random vector in the range of [0,1]; t is the current iteration; T is the maximum iteration; b is the constant defining spiral shape; l is the random number in the range [− 1, 1].

4.3 Attention-based metaheuristic CNN–BiLSTM model

The main aim of this study is to utilize metaheuristic optimization methods to improve the forecasting precision of an attention-based CNN–BiLSTM model for predicting crop yields based on historical agroclimatic and soil data. While avoiding local optima, the exploration–exploitation trade-off can be efficiently achieved by using ACO and WOA for the proposed attention-based CNN–BiLSTM model. The CNN architecture is limited to two convolutional layers to balance model complexity and generalization; hence, it reduces overfitting while maintaining crucial feature representations. The 5 layer stacked Bi-LSTM design improves the model's capacity to capture intricate temporal dynamics by learning relationships in both forward and backward directions across various time scales. Attention methods are integrated into both branches to highlight the most informative feature maps and time steps, enhancing feature relevance and interpretability. To enhance performance, ACO is utilized to optimize parameters associated with CNNs, specifically the number of convolutional filters in the Conv1D layers, while maintaining a constant kernel size ($k = 4$). Each ant signifies a proposed configuration for a CNN filter, and its fitness is assessed by validation RMSE computed on chronological hold-out validation data. In the ACO, exploration can be stimulated by evaporation of the pheromone, with an evaporation rate of $\rho = 0.2$. The evaporation process helps reduce the impact of the previously selected solutions so that the algorithm will not converge too early. Exploitation can be done by reinforcing the pheromone for the solution with a smaller RMSE. On the other hand, exploitation can be attained by using the concept of pheromone reinforcement, whereby solutions with low RMSE get more reinforcements. The probabilistic transition rule further maintains diversity in the search process. The WOA is utilized concurrently to optimize the hyperparameters of the BiLSTM branch, encompassing the amount of hidden units, dropout rate, and learning rate. Each whale encodes a candidate BiLSTM hyperparameter set, and its fitness is evaluated by validation RMSE using the same chronological hold-out strategy. In WOA, exploration and exploitation are balanced using a linearly decreasing control parameter a, which transitions from 2 to 0 over iterations. During early iterations ($|A| > 1$), the algorithm performs global exploration by updating candidate solutions relative to randomly selected whales. For the subsequent runs ($|A| < 1$), the search algorithm becomes focused on exploitation near the best solution. The optimization method employs a population of 10 ants in ACO and 12 whales in WOA for up to 20 iterations. These parameters have been chosen through initial analysis of convergence, whereby the validation RMSE is evaluated per iteration. It is observed that the algorithm converges in 15–18 iterations, while improvements are negligible after 20 iterations. A higher number of iterations or population size will result in increased computational cost without any noticeable improvement in performance. Hence, the chosen setup guarantees stability during convergence and is computationally efficient at the same time. The search space consists of CNN filters within {16, 32, 64, 96} along with BiLSTM hyperparameters including hidden layers of {32, 64, 96, 128}, dropout from {0.1–0.4}, and learning rate {0.001, 0.0005}. For the CNN and BiLSTM branch, larger designs would considerably add complexity to the model without significant gains in feature extraction on tabular time-series data. In recurrent networks, the number of learnable parameters grows very fast with hidden nodes, and overcapacity of the model with respect to the amount of training data may result in overfitting. This is why the chosen search spaces strike a balance between

model expressivity, efficiency, and generalization, allowing optimization by means of ACO and WOA. Convergence is determined based on the minimum validation RMSE, and the optimization process terminates either when the maximum number of iterations is reached or when no significant improvement in RMSE is observed over successive iterations. The optimization technique results in the proposed architecture comprising a 2 layer CNN and a 5 layer BiLSTM with an attention mechanism, as illustrated in Fig. 2, which adeptly captures intricate agroclimatic patterns and long-term temporal relationships related to crop yield variability.

Both ACO and WOA jointly minimize RMSE as showing in Eq. (23):

$$\min \text{RMSE}\left(\emptyset_{CNN}, \emptyset_{BiLSTM}\right) \tag{23}$$

where RMSE is the Root Mean Square Error; $\varnothing_{CNN}$ is the CNN parameter set; $\varnothing_{BiLSTM}$ is the BiLSTM parameterset.

CNN parameter representation using ACO in Eq. (24):

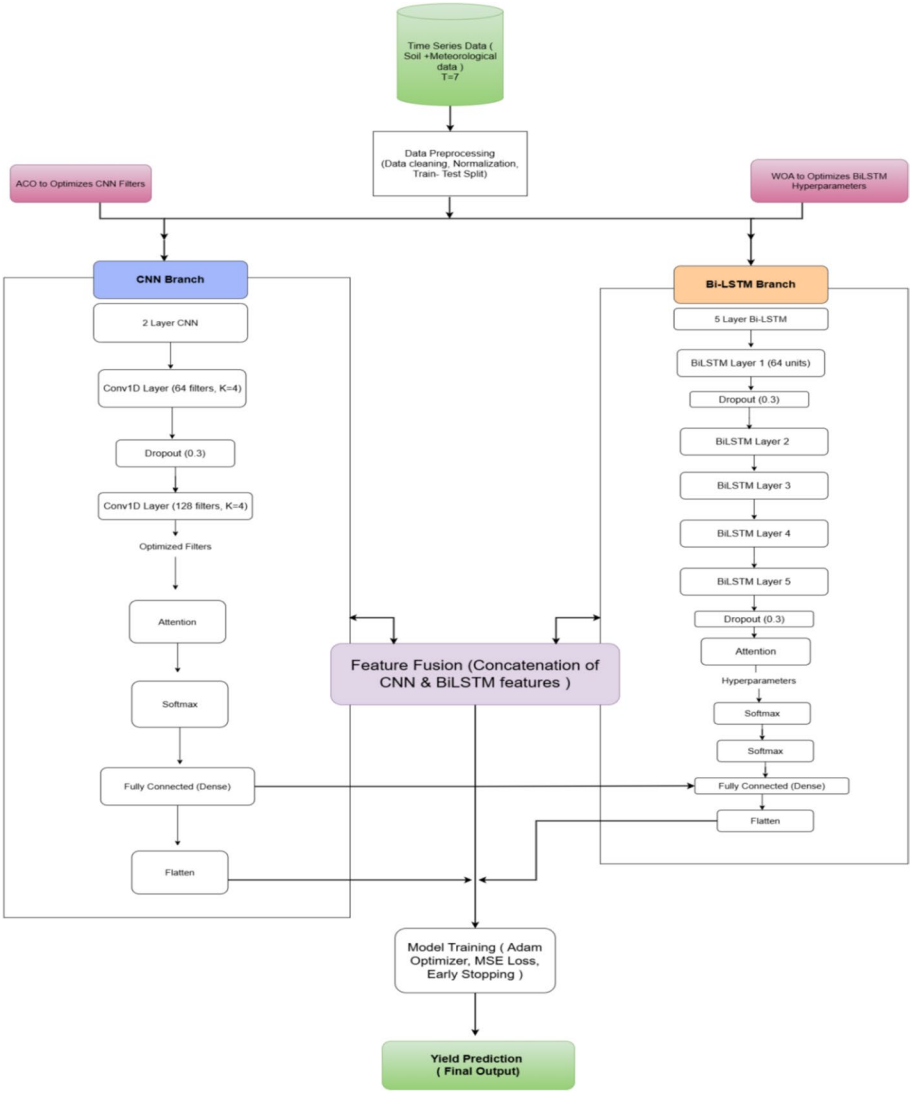


Fig. 2 Proposed attention-based metaheuristic CNN–BiLSTM model

$$\emptyset_{CNN} = \{f\} \tag{24}$$

where f is the number of convolution filters.

BiLSTM parameter representation using WOA in Eq. (25):

$$\emptyset_{LSTM} = \{h_1, h_2, h_3, h_4, h_5, d, n\} \tag{25}$$

where h_i is the hidden units of layer i; d, n is the dropout and learning rate.

Parallel hybrid optimization mapping is showing in Eq. (26):

$$\hat{y} = f\left(X \left| \emptyset_{CNN}^{ACO}, \emptyset_{BiLSTM}^{WOA} \right.\right) \tag{26}$$

where X is the input multivariate time-series data; $\widehat{y}$ is the predicted crop yield; $\varnothing_{CNN}^{ACO}$ is the CNN parameters optimized via ACO; $\varnothing_{BiLSTM}^{WOA}$ is the BiLSTM parameters optimized via WOA.

Combined optimization flow of model is showing in Eq. (27):

$$(\emptyset_{CNN}^{*}, \emptyset_{BiLSTM}^{*}) = \arg\min RMSE\left(\emptyset_{CNN}, \emptyset_{BiLSTM}\right) \tag{27}$$

where $\varnothing_{CNN}^{*}, \varnothing_{BiLSTM}^{*}$ is the optimal parameter sets.

Attention weighted final representation is represent in Eq. (28):

$$z = \left[\sum_{t=1}^{T} a_t^{CNN} c_t; \sum_{t=1}^{T} a_t^{LSTM} h_t\right] \tag{28}$$

where z is the final feature representation; c_t is the CNN feature at time t; h_t is the BiLSTM hidden state at time t; a_t^{CNN}, a_t^{LSTM} is the attention weights; T is the total time steps.

The algorithm of attention-based metaheuristic CNN–BiLSTM model is given in Algorithm 1. District-wise agro-climatic and soil dataset D containing wheat yield, soil nutrients (N, P, K, pH), and seasonal climate variables for Rajasthan (1981–2017) are the input features, and the output is predicted wheat yield.

1. Load dataset **D** from DATA.csv.
2. **For** each column **c** in dataset **D**:
   If column **c** contains missing values:
   Replace missing values with column-wise mean imputation
   End If
3. **End For**
4. Extract feature matrix **X** consisting of:
   H, W, SoilWetness, TempMax, TempMin, SpeedMax, SpeedMin, Preci, N, PH, PO, RAIN.
5. Extract target vector **y** representing wheat yield.
6. Apply Min–Max normalization to both **X** and **y**.
7. Initialize empty arrays **X_seq** and **y_seq**.
8. **For i = T** to length(X):
   X_seq[i] = X[i−T : i]
   y_seq[i] = y[i]
9. **End For**
10. Convert X_seq and y_seq into tensors.
11. Split dataset chronologically:
    Training set = first 70%
    Validation set = next 15%
    Testing set = remaining 15%
12. No random shuffling is applied. Chronological splitting is used.

13. Define Attention_Block(input)**:**
    Compute attention scores using a Dense layer
    Apply Softmax to obtain attention weights
    Multiply attention weights with input features
    Return weighted feature representation

14. Define Build_Model(cnn_filters, lstm_units, dropout, lr)
15. Define input tensor with shape (T, number_of_features).
16. Apply Conv1D(cnn_filters, kernel size = 4).
17. Apply BatchNormalization.
18. Apply Dropout (dropout = 0.3).
19. Apply Conv1D (2 × cnn_filters, kernel size = 4).
20. Apply Attention_Block.
21. Flatten output to obtain cnn_out**.**
22. **For** l = 1 to 5**:**
    Apply Bidirectional LSTM (lstm_units)
    Apply Dropout (dropout)
23. **End For**
24. Apply Attention_Block.
25. Flatten output to obtain lstm_out**.**
26. Concatenate cnn_out and lstm_out**.**
27. Apply Dense layer for feature fusion.
28. Apply final output layer to predict wheat yield.
29. Compile model using Adam optimizer and MSE loss.
30. Define fitness = RMSE between predicted and actual yield.
31. Define CNN filter candidates:
    Fc = {16, 32, 64, 96}
32. Initialize pheromone levels.
33. Set parameters:
    Population size (ants) = 10
    Maximum iterations = 20
    Evaporation rate = 0.2
34. Initialize whale population.
35. **For** each whale w**:**
    Assign random parameters:
    - LSTM units ∈ {32, 64, 96, 128}
    - Dropout ∈ {0.1, 0.2, 0.3, 0.4}
    - Learning rate ∈ {0.001, 0.0005}
36. **End For**

```
Set whale population size = 12.

For iteration = 1 to Max_iterations:
For each ant:
    Select CNN filter size using pheromone probability
    Select kernel size = 4
End For
Select whale w from whale population.
Retrieve parameters: lstm_units, dropout, learning_rate.
Build CNN–BiLSTM model using selected parameters.
Train model using training dataset.
Predict yield on validation dataset.
Compute RMSE as fitness value.
If RMSE < best_rmse:
    best_rmse = RMSE
    Store best parameters
End If
Update pheromone trails:
    pheromone = (1 − 0.2 × pheromone) + (1 / RMSE)
Compute control parameter:
    a = 2 − (2 × iteration / Max_iterations)
For each whale w:
    Generate random number r
    Compute: A = 2 × a × r − a
        If |A| < 1:
            Move whale toward best solution
        Else:
            Assign random parameters
        End If
End For
End For
Build CNN–BiLSTM model using optimized parameters.
Train final model using training and validation datasets.
Apply early stopping to avoid overfitting (patience = 20).
Predict wheat yield for test dataset.
Apply inverse normalization.
Compute performance metrics:
  RMSE, MAE, R²
Compute additional statistics:
    MAE ± Standard Deviation
    95% Confidence Interval for MAE, Paired t-test
Return results:
    Predicted yield values, RMSE, MAE, MAE ± SD, R² score, p-value
```

Algorithm 1 Attention-based metaheuristic CNN–BiLSTM model algorithm

5 Implementation

The following subsections explain the implementation details and the achieved results. The baseline methods comprise state-of-the-art techniques for crop yield prediction. The different architectural configurations evaluated in this section are part of the ablation study to analyze model depth and design choices and are not part of the final model. The final model used for evaluation is the attention-based metaheuristic CNN–BiLSTM model that has 2 layers of CNN and 5 layers of BiLSTM model.

5.1 Ablation study

An ablation study is performed to evaluate the effects of varying time-step lengths T and random seed initialization on model performance. All experiments are carried out on a standard architecture consisting of 3 convolutional layers and 3 Bi-LSTM layers without metaheuristic optimization, ensuring an unbiased and systematic comparison process. The coefficient of determination (R^2) is evaluated at three time-step value, T = 5, T = 7, and T = 9, using three different random seeds, 42, 45, and 48. For a relatively smaller

Table 2 An ablation study showing the effects of varying time-step

Time step	T = 5	T = 7	T = 9
Seed Value = 42	R2 = 0.7718	R2 = 0.8212	R2 = 0.8098
Seed Value = 45	R2 = 0.7622	R2 = 0.8182	R2 = 0.7998
Seed Value = 48	R2 = 0.6852	R2 = 0.8100	R2 = 0.7900
Mean ± SD	0.7397 ± 0.0476	0.8165 ± 0.0057	0.7999 ± 0.0099

Table 3 An ablation study showing the contribution of key architectural and optimization components

Technique	RMSE	MAE	MAE ± SD	R^2
CNN only	420.15	315.42	315.42 ± 210.33	0.7124
BiLSTM only	389.62	295.84	295.84 ± 198.56	0.7458
CNN–BiLSTM	365.78	278.11	278.11 ± 185.20	0.7685
Attention based CNN–BiLSTM model	304.05	231.77	231.77 ± 196.79	0.8212
CNN–BiLSTM-ACO	210.34	165.28	165.28 ± 120.17	0.9035
CNN–BiLSTM-WOA	198.91	152.76	152.76 ± 110.42	0.9182
Attention based CNN–BiLSTM-ACO	170.52	132.18	132.18 ± 95.31	0.9415
Attention based CNN–BiLSTM-WOA	158.74	121.63	121.63 ± 88.27	0.9528
Proposed attention based metaheuristic CNN–BiLSTM model	132.12	104.36	104.36 ± 81.02	0.9678

temporal window of T = 5, the achieved values for R^2 are 0.7718, 0.7622, and 0.6852, respectively, for random seeds 42, 45, and 48. Though the model is able to perform sufficiently well, the findings indicate limited ability to capture temporal context and heightened sensitivity to initialization. By expanding the time step to T = 7, there is a noticeable improvement in the predictive performance for all seed values. The maximum R^2 scores achieved are 0.8212 (for seed 42), 0.8182 (for seed 45), and 0.8100 (for seed 48). A slight reduction in performance is noted at T = 9, with the R2 scores being 0.8098, 0.7998, and 0.7900 for seeds 42, 45, and 48, respectively. For obtaining reliable statistical results, the findings are presented as mean ± standard deviation based on several experiments conducted with various random seeds (42, 45, and 48). In this case, the maximum accuracy is achieved at time step T = 7 with R^2 equal to 0.8165 ± 0.0057. It indicates that this experiment shows better results not only in terms of the obtained coefficient but also due to reduced variability.

Based on the findings from ablation studies, a time-step value of T = 7 using random seed 42 has been selected for further experimentation. This design has the greatest R^2 value of 0.8212 and exhibited consistent performance across several initializations, as indicated in Table 2. To ensure a fair and comprehensive evaluation, additional baseline models are implemented under identical experimental conditions. The various architectural designs discussed in this chapter are included in the ablation study to evaluate the depth and design of the model. The final model used for evaluation is the attention-based metaheuristic CNN–BiLSTM model that has 2 layers of CNN and 5 layers of BiLSTM model. To further validate the effectiveness of the proposed architecture, an ablation study is extended beyond hyperparameter sensitivity (time steps and random seeds) to systematically evaluate the contribution of key architectural and optimization components as shown in Table 3. The CNN only model produces an RMSE of 420.15, MAE of 315.42, and R2 score of 0.7124, suggesting that it is less efficient than others since it cannot capture temporal features efficiently. The BiLSTM only model performs more efficiently in terms of RMSE of 389.62, MAE of 295.84, and R2 of 0.7458 because it can efficiently model temporal relationships but cannot capture spatial features.

However, for the model without attention and optimization, which consists of CNN–BiLSTM, the RMSE is 365.78, MAE is 278.11, and R^2 is 0.7685. However, incorporating the attention model led to the improvement in the model's prediction capability as the values of RMSE is 304.05, MAE is 231.77, and R^2 is 0.8212. This indicates that attention contributes to the increased focus on important temporal features. When optimization methods are utilized separately, additional improvements are observed. For instance, the CNN–BiLSTM-ACO model produced RMSE = 210.34 and MAE = 165.28 (R^2 = 0.9035), while the CNN–BiLSTM-WOA model yielded RMSE = 198.91 and MAE = 152.76 (R^2 = 0.9182). These findings confirm the fact that optimization methods improve model accuracy greatly through optimizing parameters for a model. The combination of attention with optimization also contributed to further enhancement. The attention-based CNN–BiLSTM-ACO algorithm showed an improvement in RMSE by reducing it to 170.52 and MAE to 132.18 (R^2 = 0.9415). The attention based CNN–BiLSTM-WOA algorithm resulted in another enhancement by further reducing RMSE to 158.74 and MAE to 121.63 (R^2 = 0.9528). These findings illustrate how attention and optimization complement each other in providing improvements in the feature selection and tuning process. The proposed attention-based metaheuristic CNN–BiLSTM model performed the best with an RMSE of 132.12, MAE of 104.36, and R^2 score of 0.9678. Additionally, the model exhibited improved stability, as indicated by a lower MAE variability (± 81.02). The progressive improvement across configurations clearly shows that removing any component leads to performance degradation. The proposed model demonstrates superior performance compared to baseline and ablation models under the same dataset and experimental conditions.

5.2 Experimental setup

The experimental configuration for the proposed attention-based metaheuristic CNN–BiLSTM model is meticulously crafted to guarantee robustness, reproducibility, and realistic time-series prediction. Alternative configurations are explored during the ablation study and hyperparameter search process, and are not part of the final model. The agroclimatic dataset is initially preprocessed by addressing missing values by using column-wise mean imputation to maintain data distribution and prevent information loss. The attributes and target variables are normalized using min–max normalization to increase convergence stability when performing training using deep learning models. A sliding window approach using a time step of 7 is used to convert the data into time series datasets for capturing temporal dependence. The random seed (42) is fixed across all implementations using Python, NumPy, and Tensor Flow to achieve consistent results. The architecture comprises a 2 layer CNN (kernel size = 4) optimized via ACO for feature extraction, and a 5 layer BiLSTM optimized using the WOA for advanced temporal modeling. Dropout regularization (ideal values ranging from 0.1 to 0.4) is applied after the CNN and BiLSTM layers for avoiding overfitting, whereas attention is used for focusing on important time instances. The method of chronological hold-out validation, which consists of dividing the data into 70%, 15%, and 15% for training, validation, and testing respectively, is applied to maintain chronological order and ensure realistic prediction settings. The training procedure uses the Adam optimizer and MSE loss for up to 250 epochs using early-stopping (patience 20) to avoid overfitting. The

evaluation procedure uses RMSE, MAE, and R^2 as performance metrics to comprehensively assess the accuracy of prediction results.

6 Result and discussion

The incorporation of an optimized 2 layer CNN and 5 layer BiLSTM model into the ACO-WOA paradigm increases the accuracy of the predictions, which leads to the creation of the proposed attention-based metaheuristic CNN–BiLSTM model architecture. In the proposed framework, the ACO algorithm is employed for the optimization of CNN model parameters, while WOA is used for optimizing the BiLSTM hyperparameters. The optimization process is performed prior to training the final model. This is because it will be easier for the CNN model to learn the short-range spatial and temporal features from the agroclimatic data through its structure, whereas the BiLSTM is capable of learning the long-range temporal features such as seasonality and variations between years. This allows the model to accurately simulate the growth pattern of the crops, hence enhancing prediction accuracy. Optimization begins with the initialization of the population. In terms of the CNN part, the important aspect that is to be optimized using the ACO algorithm is the number of convolutions in the first Conv1D layer, chosen among values {16, 32, 64, 96}. After determining the optimal value, the next convolution layer will use double the number of filters ($2 \times$ cnn_filters). The kernel size should remain constant at 4. Use ReLU activation along with same padding in order not to lose temporal resolution. The value of the dropout is chosen as 0.3 to mitigate overfitting. An attention block is applied after the second convolution layer to increase the importance of the features. The attention block makes use of learnable dense layers using the tanh activation function to calculate the attention weights. Then, the block applies a softmax function on the temporal dimension. The attention mechanism helps improve the performance of the model by allowing the network to concentrate on more informative time points of the crop growth process minimizing irrelevant inputs and enhancing feature discrimination. The attention-weighted CNN features are subsequently flattened and passed through a dense layer in order to give a unique feature representation from the CNN perspective. In the case of the BiLSTM, the number of hidden nodes per LSTM layer is determined from {32, 64, 96, 128}. Furthermore, the dropout rate is selected from {0.1, 0.2, 0.3, 0.4} in order to address issues relating to overfitting, whereas the learning rate is set from {0.001, 0.0005} as shown in Table 4. The ACO algorithm

Table 4 Parameter settings for the proposed model

Optimization algorithm	Model component	Optimized parameter	Search range/ candidate values	Description
ACO (Ant Colony Optimization)	CNN Branch	Number of CNN filters	{16, 32, 64, 96}	Determines the number of convolutional filters used in the first Conv1D layer for spatial–temporal feature extraction
	CNN Branch	Number of filters in second Conv layer	$2 \times$ cnn_filters	Automatically determined based on the first layer filter size
WOA (Whale Optimization Algorithm)	BiLSTM Branch	LSTM Hidden Units	{32, 64, 96, 128}	Determines the number of neurons in each Bidirectional LSTM layer
	BiLSTM Branch	Dropout Rate	[0.1–0.4]	Prevents overfitting by randomly dropping neurons during training
	BiLSTM Branch	Learning Rate	{0.001, 0.0005}	Controls the optimization step size during BiLSTM training

initialized a population of 10 ants with equal pheromone levels corresponding to the CNN filter choices. Simultaneously, the WOA algorithm initialized a population of 12 whales, where each whale represented a candidate solution consisting of LSTM units for the five BiLSTM layers along with a dropout rate and learning rate. The pheromone evaporation rate in ACO is set to 0.2 to balance exploration and exploitation during the search process. The optimization procedure is executed for 20 iterations, during which the best solution is continuously tracked based on the minimum RMSE value obtained. In the proposed framework, a fitness function is used by both ACO and WOA to test candidate solutions. The fitness function measures how well the model predicts the validation dataset. The RMSE between the predicted and actual yield values is used to figure out the fitness value of each candidate solution, as shown in Eq. (29). A lower RMSE means that the prediction is more accurate, which means that the solution is better [25]. The goal of the optimization process is to minimize the prediction error.

$$\text{Fitness} = \text{RMSE} = \sqrt{\frac{1}{m}\sum_{i=1}^{m}(y_i - \hat{y}_i)^2} \tag{29}$$

where y_i is the actual crop yield; $\widehat{y}_i$ is the predicted crop yield; m is the number of samples.

The next step is to update the parameters after calculating the fitness values. In ACO, the RMSE results are used to change the pheromone values linked to CNN filter configurations. This makes it more likely that high-performing configurations will be chosen in subsequent iterations. The optimization process goes on until the stopping condition is reached. The stopping criteria consist of achieving a predetermined number of optimization iterations or observing no additional enhancement in the optimal RMSE value. The optimal configuration for the final CNN–BiLSTM model is the parameter set that result in the lowest RMSE during the optimization process.

The iterative improvement process makes sure that the model gradually converges to an optimal design, providing more stable predictions and less variance than non-optimal models. The Adam optimizer is used to train the proposed model, and the mean squared error (MSE) is used as the loss function. The network can be trained for up to 250 epochs, with a batch size of 32. An early stopping mechanism with a patience of 20 epochs is used to prevent overfitting and make the model more general. This mechanism restores the best model weights when the validation performance stops improving. To keep the time-series data's temporal causality, the dataset is split into three parts based on time. The first 70% of the sequences are used for training, the 15% for validation, and the last 15% for testing, with no random shuffling. The feature representations from the CNN and BiLSTM branches are combined to make a single feature vector that combines short-term spatial–temporal patterns with long-term bidirectional temporal correlations. The combined representation goes through a fully connected layer that uses ReLU activation to capture higher-order nonlinear interactions. A single neuron linear output layer makes a continuous prediction of agricultural yield. It is the integration of spatial, temporal, and enhanced attention-based elements, along with the choice of optimal parameters, that makes the model perform better than other approaches. It should be noted that all performance gains are measured on the same data using the same

experimental framework to ensure a fair comparison against other baseline and ablation models.

7 Performance evaluation

It is important to note that the evaluation is based on a single chronological split rather than a full walk-forward validation. This approach preserves temporal consistency and avoids data leakage; future work may incorporate rolling window (walk-forward) validation to further strengthen the robustness and generalizability of the evaluation. The performance measures will be calculated after applying the inverse transform on the normalized values of the target variable. The optimized model runs simultaneously three CNN layers and three BiLSTM layers. Consequently, the R^2 value is determined as 0.8990, thus providing a considerable improvement compared to the non-optimized baseline model. In order to explore the effects of temporal analysis on the BiLSTM part of the architecture, add more BiLSTM layers to the model. The 3 layer CNN-4 layer BiLSTM configuration achieved an R^2 score of 0.8751, whereas the 3 layer CNN-5 layer BiLSTM configuration performed better with a score of 0.9379. This indicates that more profound bidirectional temporal representations are superior. Consequently, the number of convolutional layers in the CNN branch is diminished to assess the impact of feature abstraction complexity inside the parallel architecture. The model, comprising 2 layer CNN-3 layer BiLSTM, attained an R^2 value of 0.9545, surpassing all configurations utilizing 3 layers CNN. By augmenting the BiLSTM depth within the parallel framework, the 2 layer CNN-4 layer BiLSTM model achieved an R^2 score of 0.9458, while the 2 layer CNN-5 layer BiLSTM configuration yielded the optimal predictive performance with an R^2 value of 0.9678. The final proposed model, as indicated in Table 5, has an attention based metaheuristic CNN–BiLSTM model with a 2 layer CNN branch with a 5 layer BiLSTM branch, trained with $T = 7$ and seed = 42. This configuration provides the optimal balance among spatial–temporal feature extraction, long-term bidirectional dependence modeling, and generalization capacity; thus, it is chosen as the final optimized model.

As can be seen from the experimental results, the proposed model performs considerably better than baseline models. The R^2 of the baseline attention-based CNN–BiLSTM model is 0.8212. Thus, indicating that this architecture has good capabilities to learn both spatial and temporal dependencies. Nonetheless, the high RMSE of 304.05 and MAE of 231.77 demonstrate some weaknesses in generalizability and sensitivity to data variability. The improvement in performance achieved by the proposed model can be attributed to the synergistic effect of different architecture components within it. While the CNN model succeeds in identifying localized spatial and temporal characteristics of agroclimatic data, such as variations in rainfall or temperature, the BiLSTM model aids in recognizing temporal dependencies in the long term, which include seasonal variations along with interannual variations. This approach ensures a more complete description of crop growth behavior, which enhances prediction accuracy. The introduction

Table 5 CNN and BiLSTM configuration with different layers

R2 score	2 layer CNN	3 layer CNN
3 layer BiLSTM	0.9545	0.8990
4 layer BiLSTM	0.9458	0.8751
5 layer BiLSTM	0.9678	0.9379

Table 6 Performance Evaluation of Baseline and proposed model

Technique	RMSE	MAE	MAE ± SD	MAE (95% CI)	R^2
Attention-based CNN–BiLSTM model	304.0529	231.7733	231.7733 ± 196.7976	[220.9494, 242.3204]	0.8212
Proposed attention-based metaheuristic CNN–BiLSTM model	132.1194	104.3635	104.3635 ± 81.0172	[99.3924, 109.4469]	0.9678

Table 7 Statistical significance analysis using paired t-test

Parameter	Value
Test Type	Paired t-test
Sample Size (n)	100
t-statistic	− 3.209
p-value	0.0018
Result	Statistically Significant Improvement

of the attention component improves the efficiency of the algorithm by placing greater emphasis on the significant times during crop growth. This ensures that the model concentrates on the important time frames that influence yield, thus decreasing the impact of less important features. One major reason why there is an increase in effectiveness is the use of metaheuristic optimization algorithms. The ACO algorithm boosts the efficiency of CNN by providing the best filter combination, while the WOA algorithm enhances temporal modeling by tuning BiLSTM hyperparameters. In contrast to the single-optimizer approaches, the proposed branch-specific optimization ensures an effective trade-off between exploration and exploitation for varying parameters, leading to improved convergence and escaping local minima. Thus, the proposed model achieves RMSE = 132.12, MAE = 104.36, and $R^2 = 0.9678$, which is significantly superior to the baseline model, as illustrated in Table 6. In addition, the decrease in MAE standard deviation from 231.77 ± 196.79 to 104.36 ± 81.02 also suggests improved stability and reliability of the optimized model. To further establish the importance of the aforementioned improvement, a paired t-test was done on the prediction errors of the proposed model and the baseline model. As seen in the results, there is a significant difference between the two ($t = -3.209$, $p = 0.0018$), with the p-value being less than 0.05, indicating that there is statistical significance at the 95% confidence level. The statistical significance of the improvement is shown in Table 7. It proves that the performance improvement is not by chance but because of the introduction of the hybrid feature learning method and others like attention weighting and branch optimization. All these factors contribute towards minimizing prediction error and generalization; thus, demonstrating the effectiveness of the proposed model.

The parallel optimization using ACO and WOA identified the optimal hyperparameters for the hybrid CNN–BiLSTM model after completing the defined iterations. The ACO algorithm selected 64 convolutional filters for the CNN layers as the best configuration based on the minimum RMSE achieved during optimization. Simultaneously, the WOA algorithm determined the optimal number of hidden units for the five BiLSTM layers as 32, 128, 32, 128, and 32, respectively. In addition to the network architecture parameters, the optimization process selected a dropout rate of 0.3 to improve generalization and a learning rate of 0.0005 for stable training of the model, as shown in Table 8. These hyperparameters are then used to train the model in order to predict wheat crop yields.

Table 8 Best hyperparameters evaluated by model

Parameters	Values
CNN Filters	64
BiLSTM Layer1	32
BiLSTM Layer2	128
BiLSTM Layer3	32
BiLSTM Layer4	128
BiLSTM Layer5	32
Dropout	0.3
Learning Rate	0.0005

Figure 3 shows the performance evaluation and diagnostic plots employed in assessing the effectiveness of the proposed model. The Table 9 shows the actual and predicted values of wheat crop yield. The loss function graph shows faster convergence in early epochs, followed by a stable learning process, indicating effective learning without overfitting. The plot of the residual map of the first 100 sample points shows that there is an equal spread of error from zero, implying that there are no predictable variations in the results of unbiased predictions. The graph showing the comparison of actual and expected yields and the 95% confidence interval suggests that predicted values closely match the actual yields, with most of the observations within the range of the confidence interval, thus suggesting the reliability of the predictive power of the model. The scatter plots of expected yield against residuals and actual yield against residuals show random scattering of points about zero, which suggests that there is homoscedasticity of errors and uniformity of model performance regardless of yield values. The residual histogram follows a normal distribution centered on zero, indicating little to no bias in the model, where most of the predictions made have small errors, with hardly any large deviations present. Correlation plots show a strong positive relationship between soil moisture and precipitation and nutrients and yield, thus proving that both are essential factors when determining crop yields. An inverse relationship between temperature, wind speed, and yield demonstrates that high temperatures and wind speeds can reduce the efficiency of agriculture production. These findings prove the effectiveness, reliability, and adaptability of the proposed model in estimating crop yields.

The comparative performance of the baseline attention based CNN–BiLSTM model and proposed attention-based metaheuristic CNN–BiLSTM model is depicted in the grouped bar plot with error bars, as illustrated in Fig. 4. The proposed model demonstrates a significant decrease in prediction error, with RMSE declining from 304.05 to 132.12, indicating an enhancement of 56.54%. Likewise, MAE decreases from 231.77 to 104.36, reflecting a 54.97% enhancement. Alongside the reduction of errors, the coefficient of determination (R^2) rises from 82.12% to 96.78%, reflecting a 17.83% increase in explained variance.

7.1 Comparative analysis

Table 10 provides a comparative analysis of the proposed models compared to the existing models for crop yield prediction. It is important to note that comparisons with existing studies are indicative rather than directly comparable, as different datasets, crops, and experimental settings are used. Therefore, the superiority of the proposed model is primarily validated through consistent improvements over baseline models under the same dataset and experimental conditions. In the research by Li et al. [14], the

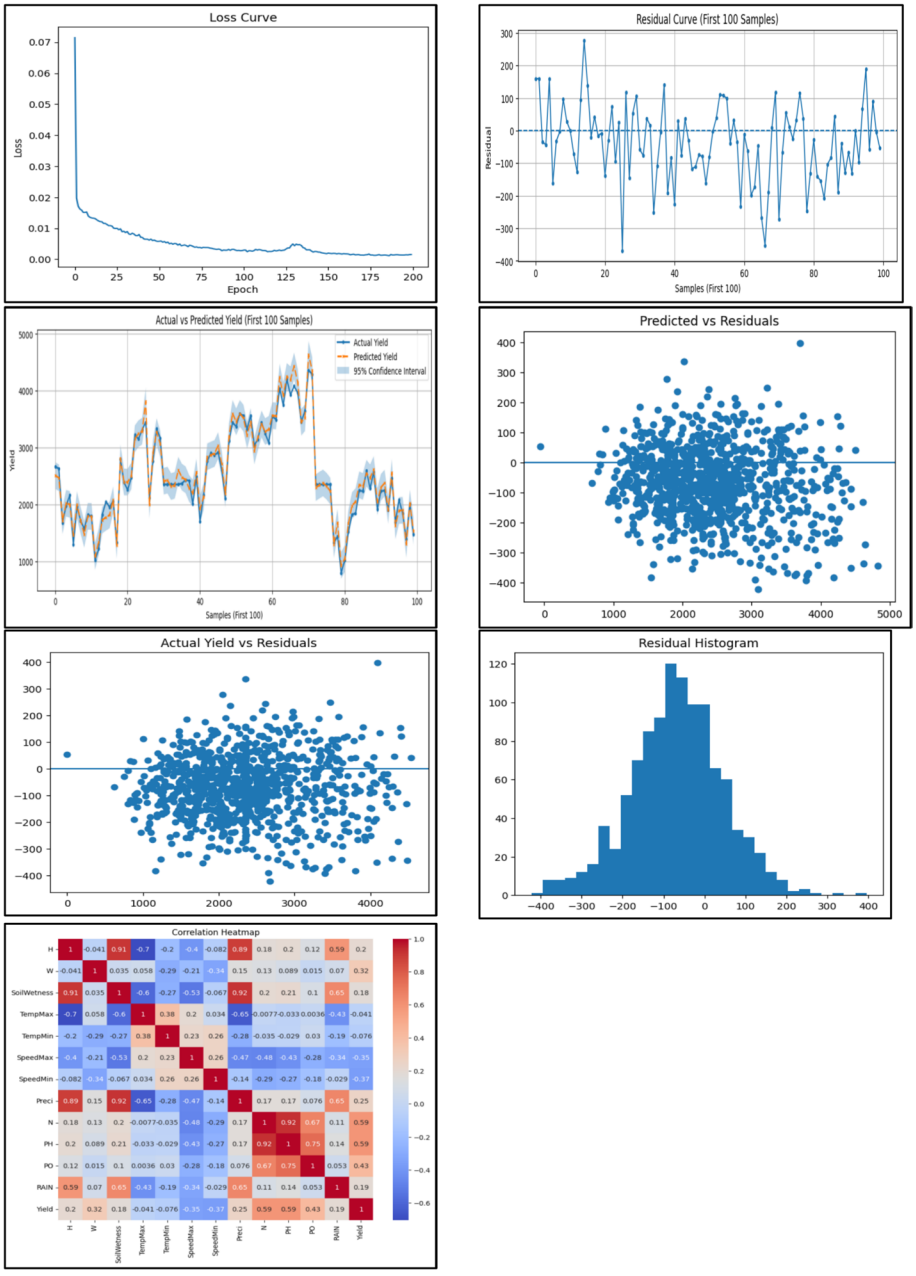


Fig. 3 Performance evaluation and diagnostic plots of the proposed model

researchers used the attention LSTM algorithm along with Sentinel-2 remote sensing data. Although the researchers obtained moderate accuracy ($R^2 \approx 0.88$), the absence of optimization resulted in the inability to obtain the optimal accuracy. In the study by Sun et al. [15], the researchers improved the prediction accuracy by using the WT-CNN–BiLSTM model, obtaining $R^2 \approx 0.95$. The TPCLA model developed by Song et al. [16] used transfer learning and attention mechanisms. This model also obtained excellent results with $R^2 \approx 0.904$ and low values of RMSE and MAE. Although this model is robust, it only uses a single optimization technique. Moreover, tuning of individual components of this model is also not done. In the same context, the model developed in the study proposed by Zhang et al. [3] is also improved using hyperparameter tuning.

Table 9 Actual and predicted wheat yield obtained from proposed attention-based metaheuristic CNN–BiLSTM model

Actual yield	Predicted yield
2667.82	2509.62
2631.40	2471.17
1684.16	1719.10
1979.66	2022.52
2162.52	2003.28
1301.97	1463.56
1963.01	1994.71
1718.33	1721.67
1575.44	1478.49
1821.13	1793.28

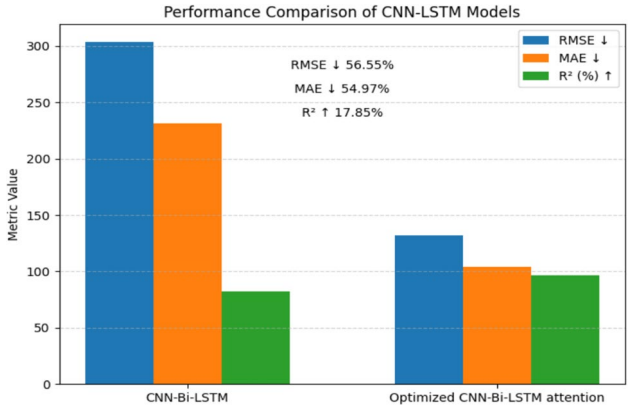


Fig. 4 Performance comparison of baseline model and proposed model

Although this model has relatively low accuracy ($R^2 \approx 0.81$), it is limited to using only a single optimization technique. On the other hand, it can be seen that the proposed model outperforms the baseline and internally evaluated models, whereas it shows stable performance when compared with existing models. By making it possible to optimize different parts of the CNN and BiLSTM models independently and by striking a proper balance between exploration and exploitation using two different metaheuristic optimization algorithms, the proposed model has shown promising performance in terms of higher R^2 value (0.9678) and significantly lower error values (RMSE = 132.12, MAE = 104.36). Moreover, using long-term multivariate agroclimatic and soil information will also improve the robustness and generality of the model. Thus, not only is the proposed approach overcoming the limitations of existing approaches, but it is also more accurate and reliable for crop yield prediction. However, caution is advised when making comparisons across studies due to the variations in databases, crops, regions, and evaluation methods.

8 Conclusion

This study shows how to use agroclimatic and soil nutrient time-series data together to make a more advanced deep learning framework for predicting wheat yield. Unlike most existing studies that employ hybrid architectures such as CNN–LSTM or CNN–BiLSTM, the proposed model introduces an attention based metaheuristic CNN–BiLSTM model for more effective feature extraction and temporal learning. A key novelty of this

Table 10 Comparative analysis of proposed model with existing literature

Existing study	Region	Feature types	Dataset size	Models tested	Computational cost	Best model & performance	Limitations
Zhang et al. [3]	China	Multi-source data (remote sensing + agro-climatic features)	~1200	CNN, BiLSTM, CNN–BiLSTM (BO optimized)	High	CNN–BiLSTM (BO): Improved RMSE = 616.99, MRE = 7.14% and R2 = 0.81	Single optimization; no independent tuning
Li et al. [14]	India	Sentinel-2 remote sensing data (vegetation indices such as NDRE, GCVI, NDWI) collected over multiple crop growth stages	~800	RF, SVR, LSTM, Attention-LSTM (ALSTM)	Moderate	ALSTM: RMSE ≈ 341.82 kg/ha, MAE ≈ 280.29 kg/ha, $R^2 \approx 0.88$	No optimization
Sun et al. [15]	Yunnan	WT-CNN–BiLSTM	~800	CNN, BiLSTM, WT–CNN–BiLSTM	Moderate–High	WT–CNN–BiLSTM achieved $R^2 \approx 0.95$ with improved forecasting accuracy	No metaheuristic optimization
Song et. al [16]	China	Multivariate wheat dataset (1993-2024), climatic + socio-economic data	~900	RNN, LSTM, LSTM-Attention, PCLA, TPCLA	High	TPCLA: RMSE = 0.394, MAE = 0.326, R2 = 0.904	Uses single optimization
Proposed study (2026)	1221	Multivariate agro-climatic and soil time-series data (rainfall, temperature, soil moisture, nutrients, etc.)	India (Rajasthan)	CNN-LSTM, CNN-LSTM with optimization, CNN–BiLSTM with ACO-WOA optimization	Moderate–High	CNN–BiLSTM (Optimized): RMSE = 132.12, MAE = 104.36, MAE ± SD = 104.36 ± 81.02, MAE (95% CI) = [99.39, 109.45], $R^2 = 0.9678$	Addresses independent optimization and improves prediction accuracy (Dual optimization)

work is the independent branch-specific optimization of the CNN and BiLSTM components using two different metaheuristic algorithms, where ACO is used to optimize the CNN parameters and WOA is used to optimize the BiLSTM hyperparameters. To the best of our knowledge, existing studies have not explicitly investigated such dual, branch-specific optimization within a unified crop yield prediction framework, which addresses the limitations of single-optimizer approaches in handling heterogeneous parameter spaces. Experimental results indicate that the proposed model attains higher accuracy and better generalization compared to baseline and internally evaluated models

with an RMSE of 132.12, an MAE of 104.36 ± 81.02, a narrow 95% confidence interval of [99.39, 109.45], and a significantly higher R2 value of 0.9678. The findings reveal that both the attention-based mechanism and the optimization methods contribute substantially to improving the accuracy, stability, and generalization of predictions. Thus, the proposed model offers a novel solution for predicting crop yields by leveraging agro-climatic information from multiple sources. However, it should be mentioned that making strict comparisons with prior research can be inappropriate owing to inconsistencies in the datasets, locations, and testing environments used by the researchers. Future study may enhance the proposed model by integrating high-resolution remote sensing data and real-time satellite observations to improve prediction accuracy. The framework may be expanded to include additional crops and regions to evaluate its generalization capacity. Also, adding advanced architectures such as graph neural networks or transformer-based temporal models could enhance spatio-temporal crop yield prediction even better.

9 Limitations

Although the model works well, there are some limitations that need to be addressed. For instance, the model is completely based on the historical agroclimatic and soil data collected, and this may fail to account for the real-time environmental changes. Moreover, there is no incorporation of real-time data into the model. Additionally, the study uses data from a particular area called Rajasthan, which might impact the applicability of the model to other areas due to differences in climatic conditions and soil type. Though the architecture can be modified, more validations should be carried out in different geographical locations. The application of metaheuristic optimization algorithms (ACO and WOA) makes the system even more complex and requires longer training times than traditional approaches. Though there is no issue with the higher accuracy provided, it might affect scalability in resource-limited settings. Additionally, the evaluation of the model has yet to be performed under a real-time deployment environment, which is crucial for making agricultural decisions.

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Author contributions
K. P.—Conceptualization of the study, design and development of the deep learning models, data pre-processing framework, experimental setup, and manuscript preparation. V. B.—Implementation of optimization algorithms for model enhancement, performance evaluation, and comparative analysis. P. K. T.— Visualization of results, preparation of figures and tables, manuscript formatting, and proofreading.

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Data availability
The datasets generated and/or analysed during the current study are available in the ICRISAT and NASA POWER repository, (https://www.icrisat.org/resources/data/comparative-map-homepage, https://power.larc.nasa.gov/data-access-viewer/).

Declarations

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Not applicable.

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The authors declare no competing interests.

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